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RFID tag, blood collection tube, and antenna

Abstract

The present invention provides an RFID tag to be applied to a container for holding liquid. To achieve a good communication distance regardless of the presence or absence of liquid, the RFID is applied to a container for holding liquid, and includes an IC chip on which identification information is recorded, and an antenna made of a loop-shaped conductor connected to the IC chip, and the antenna has a T-shaped opening in which the conductor is not formed.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to an RFID (Radio Frequency Identification) tag, a blood collection tube, and an antenna.
BACKGROUND ART
(2) RFID labels to be applied to objects are popular in logistics management, product management, and in other fields. An RFID label is a label with an RFID tag. An RFID tag includes an IC chip and an antenna that is electrically connected to the IC chip. An RFID tag may be also referred to as a “wireless tag,” an “IC tag,” an “RF-ID tag,” an “RF tag,” and the like.
(3) Also, as for the antenna for RFID tags, there is an antenna formed with a power-feeding terminal to which an IC chip is connected, and a bypass conducting path that bypasses the loop of a loop antenna (see, for example, Patent Documents 1 and 2).
RELATED-ART DOCUMENTS
Patent Documents
(4) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2007-19905
Patent Document 2: International Publication No. WO2006/077645
SUMMARY OF THE INVENTION
Problem to be Solved by the Invention
(5) Patent Documents 1 and 2 disclose that a good communication distance can be achieved in a frequency band in which an RFID tag communicates wirelessly, by making a capacitance component inside an IC chip and an inductance component of a bypass conducting path provided in an antenna resonate.
(6) However, with RFID tags to be applied to containers for holding liquid, such as blood collection tubes used for blood tests, there is a problem that the impedance characteristics of the antenna change when the container holds liquid, and the communication distance is shortened.
(7) One embodiment of the present invention has been made in view of the above, and provides an RFID tag that is applied to a container for holding liquid, and that can achieve a good communication distance regardless of the presence or absence of liquid.
Means to Solve the Problem
(8) In order to solve the above problem, one embodiment of the present invention provides an RFID tag to be applied to a container for holding liquid, the RFID tag including: an IC chip on which identification information is recorded; and an antenna made of a loop-shaped conductor connected to the IC chip, and, in this RFID tag, the antenna has a T-shaped opening in which the

conductor is not formed.

Effects of the Invention

(9) According to one embodiment of the present invention, it is possible to provide an RFID tag that is applied to a container for holding liquid and that achieves a good communication distance regardless of the presence or absence of liquid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram that illustrates an example appearance of a blood collection tube to which an RFID tag according to one embodiment is applied;

(2) FIG. **2** is a diagram that illustrates an example structure of an RFID tag antenna according to one embodiment;

(3) FIG. **3** is a diagram that illustrates the shape of an antenna according to one embodiment;

(4) FIG. **4** is a diagram that illustrates an example of impedance characteristics of an antenna according to one embodiment;

(5) FIG. **5** is a diagram that illustrates an example of the shape of an antenna according to a comparative example;

(6) FIG. **6** is a diagram that illustrates an example of impedance characteristics of an antenna according to a comparative example;

(7) FIG. **7** is a diagram that illustrates an example of the shape of an antenna according to a first modification;

(8) FIG. **8** is a diagram that illustrates an example of impedance characteristics of the antenna according to the first modification;

(9) FIG. **9** is a diagram that illustrates an example of the shape of an antenna according to a second modification;

(10) FIG. **10** is a diagram that illustrates an example of impedance characteristics of the antenna according to the second modification;

(11) FIG. **11** is a diagram that illustrates an example of the shape of an antenna according to a third modification;

(12) FIG. **12** is a diagram that illustrates an example of impedance characteristics of the antenna according to the third modification;

(13) FIG. **13** is a diagram that illustrates an example of the shape of an antenna according to a fourth modification;

(14) FIG. **14** is a diagram that illustrates an example of impedance characteristics of the antenna according to the fourth modification;

(15) FIG. **15** is a diagram that illustrates an example of the shape of an antenna according to a fifth modification;

(16) FIG. **16** is a diagram that illustrates an example of impedance characteristics of the antenna according to the fifth modification;

(17) FIG. **17** is a diagram that illustrates an example of the shape of an antenna according to a sixth modification;

(18) FIG. **18** is a diagram that illustrates an example of impedance characteristics of the antenna according to the sixth modification;

(19) FIG. **19** is a diagram that illustrates an example arrangement pattern of blood collection tubes according to one embodiment; and

(20) FIG. **20** is a diagram that illustrates an example of an RFID tag evaluation result according to one embodiment.

MODE FOR CARRYING OUT THE INVENTION

(21) Now, an embodiment of the present invention will be described in detail with reference to the accompanying drawings. In the description below, parts that are the same in each drawing may be assigned the same reference numerals, and their description may be omitted. Also, for ease of understanding, the scale of each member in each drawing may differ from the actual scale. Note that, in each example illustrated, directions that are referred to by words such as "parallel," "orthogonal," "horizontal," "vertical," "up," "down," "left," "right," and so forth may tolerate minor deviation insofar as such deviation does not mitigate the effects of the present invention. Also, the X-axis direction, the Y-axis direction, and the Z-axis direction include directions that are parallel to the X-axis, directions that are parallel to the Y-axis, and directions that are parallel to the Z-axis. The X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to each other. The XY plane, the YZ plane, and the ZX plane include virtual planes that are parallel to the X-axis direction and the Y-axis direction, virtual planes that are parallel to the Y-axis direction and the Z-axis direction, and virtual planes that are parallel to the Z-axis direction and the X-axis direction, respectively. Of the directions than run along the X axis, the direction indicated by the arrow is the plus X-axis direction, and the opposite direction is the minus X-axis direction. Of the directions that run along the Y axis, the direction indicated by the arrow is the positive Y-axis direction, and the opposite direction is the negative Y-axis direction. Of the directions that run along the Z axis, the direction indicated by the arrow is the positive Z-axis direction, and the opposite direction is the negative Z-axis direction.
(22) <Appearance of Blood Collection Tube>
(23) FIG. **1** is a diagram that illustrates an example appearance of a blood collection tube to which an RFID tag according to one embodiment is applied. The example of FIG. **1** shows the appearance of a blood collection tube **10**, which is an example of a container for holding liquid. An RFID tag **100** is applied to the blood collection tube **10** in a predetermined application range **12** on the outer periphery. Preferably, a cap **11** is provided at the tip of the blood collection tube **10**.
(24) The blood collection tube **10** is made of plastic such as polyethylene terephthalate. However, the blood collection tube **10** is by no means limited to this, and may be made of other materials such as glass.
(25) Note that the blood collection tube **10** is an example of a container for holding liquid, and a container other than the blood collection tube **10**, such as a test tube, a urine collection tube, or an ampoule, may be used as well. Also, blood is an example of liquid that the container holds, and liquids other than blood such as urine, beverage, medicine, and water, may be used as well. Here, an example will be described below assuming that the blood collection tube **10** is the container for holding liquid.
(26) <Structure of RFID Tag>
(27) FIG. **2** is a diagram that illustrates an example structure of an RFID tag according to one embodiment. The RFID tag **100** includes, for example, a band-like sheet **200**, an IC chip **210** on which identification information is recorded, an antenna **220** made of a loop-shaped conductor connected to the IC chip **210**, and so forth. Also, the inner periphery **221** of the antenna **220** has a shape like the letter T.
(28) The sheet **200** and the antenna **220**, for example, have outer shapes that match the blood collection tube **10**, as shown in FIG. **1**. For example, it is desirable to make the depth D0 of the sheet **200** and the depth D1 of the antenna **220** shorter than the minimum value of the perimeter in the predetermined application range **12** of the blood collection tube **10**. As a result of this, for example, as shown in FIG. **1**, the RFID tag **100** can be affixed to the blood collection tube **10** such that the RFID tag **100** does not overlap itself. Also, it is desirable to keep the width W0 of the sheet **200** and the width W1 of the antenna **220** within the predetermined application range **12** of the blood collection tube **10**.
(29) Note that the width W1 and the depth D1 of the antenna **220** can be determined flexibly within the above ranges. For example, if the frequency that the RFID tag **100** uses is 920 MHz, W1 can be

20.00 mm to 50.00 mm, preferably 30.00 mm to 40.00 mm. Also, D1 can be 13.00 mm to 18.00 mm, preferably 15.00 mm to 17.00 mm.

(30) The sheet **200** is, for example, a band-like film formed by laminating a plurality of synthetic resin films such as polyethylene terephthalate and polypropylene. However, the sheet **200** is by no means limited to this, and the sheet **200** may be made of paper or the like. The IC chip **210** and the antenna **220** are arranged so as to be, for example, sandwiched between a plurality of laminated synthetic resin films. Note that the RFID tag **100** may be provided in a label that is applied to the blood collection tube **10**.

(31) The IC chip **210** is an REIN system integrated circuit on which identification information is recorded, and is electrically connected to the antenna **220**. The IC chip **210** receives, via the antenna **220**, radio waves transmitted at a predetermined radio frequency (for example, 920 MHz band: 860 MHz to 960 MHz) from the tag reader of the RFID system, generates power from the received radio waves, and is activated. Also, the IC chip **210** uses the generated power to transmit radio waves containing identification information pre-recorded in the IC chip **104**, to the tag reader.

(32) The antenna **220** is made of a highly conductive conductor (for example, a metal such as copper or aluminum) in a loop shape. Also, the inner periphery **221** of the antenna **220** is shaped like the letter T, and the conductor is not formed in this inner periphery **221**. In this embodiment, this part inside the inner periphery **221** where no conductor is formed is referred to as an “opening **222**.” Thus, the antenna **220** is made of a loop-shaped conductor connected to the IC chip **210**, and has a T-shaped opening **222** in which no conductor is formed.

(33) Preferably, the outer shape of the antenna **220** is a rectangle with a width of W1 and a depth of D1, as shown in FIG. **2**. Also, the opening **222** includes: a first opening **223** that extends in a direction (X-axis direction) parallel to one side of the rectangle on which the IC chip **210** is mounted; and a second opening **224** that is connected to a central part of the first opening **223**, and that extends in a direction perpendicular to one side of the rectangle on which the IC chip **210** is mounted (minus Y-axis direction). Note that, in FIG. **2**, the dashed two-dotted line, representing the boundary between the first opening **223** and the second opening **224**, is an imaginary line and does not really exist.

(34) Preferably, the RFID tag **100** is applied to the blood collection tube **10** such that the longitudinal direction of the blood collection tube **10** and the direction in which the first opening **223** extends (X-axis direction) are substantially parallel. By this means, better communication characteristics can be achieved. Note that, when the RFID tag **100** is applied to the blood collection tube **10**, there is no significant difference in communication characteristics between the case in which the plus X direction of the RFID tag **100** shown in FIG. **2** faces the cap **11** of the blood collection tube **10**, and the case in which the minus X direction of the RFID tag **100** shown in FIG. **2** faces the cap **11** of the blood collection tube **10**.

(35) Note that the RFID tag **100** may be applied to the blood collection tube **10** such that, for example, the longitudinal direction of the blood collection tube **10** and the direction in which the first opening **223** extends cross.

(36) Note that the width W2 and depth D2 of the first opening **223** and the width W3 and depth D3 of the second opening **224** can be determined flexibly. For example, when the frequency the RFID tag **100** uses is 920 MHz, W2 can be 18.00 mm to 25.00 mm, preferably 20.00 mm to 23.00 mm, more preferably 21.10 mm to 22.10 mm. D2 can be 1.00 mm to 7.00 mm, preferably 3.00 mm to 5.00 mm, more preferably 3.50 mm to 4.50 mm. W3 can be 1.00 mm to 7.00 mm, preferably 3.00 mm to 5.00 mm, more preferably 3.7 mm to 4.7 mm. D3 can be 3.00 mm to 9.00 mm, preferably 5.00 mm to 7.00 mm, more preferably 5.50 mm to 6.5 mm.

(37) The antenna **220** is made of a conductor such as a metal foil like copper or aluminum that is pressed, etched, or plated, a metal paste that is silk-screen-printed, and so forth. Note that, if the conductor is aluminum, the thickness of the conductor can be, for example, 5 μm to 40 μm, preferably 7 μm to 30 μm. Also, although the antenna **220** is hatched in FIG. **2** and others, this

hatching only indicates that the antenna **220** is made of metal, and is not a pattern.
(38) In the above structure, the IC chip **210** has internal capacitance, and this internal capacitance and the inductance component of the antenna **220** form a resonant circuit (matching circuit). In this resonant circuit, the imaginary component becomes practically zero at the resonance frequency at which the internal capacitance of the IC chip **210** and the inductance component of the antenna **220** resonate, thereby achieving impedance matching and ensuring a sufficient communication distance.
(39) The RFID tag **100** is thus configured to achieve a good communication distance regardless of the presence or absence of liquid (for example, blood) in the blood collection tube **10** at, for example, a frequency in the 920 MHz band (860 MHz to 960 MHz, preferably 915 MHz to 935 MHz).
(40) <The Shape and Impedance Characteristics of the Antenna>
(41) FIG. **3** is a diagram that illustrates the shape of the antenna according to one embodiment. This figure illustrates an example of the shape of the antenna **220** with which good communication characteristics have been achieved. Referring to FIG. **3**, the width W1 of the antenna **220** was made 35.00 mm and the depth D1 was made 16.00 mm, and the opening **222** was formed so as to be symmetrical with respect to the center line **225**. Note that the symmetrical shape tolerates error and deviation insofar as such error and deviation do not mitigate the effect of the present invention. Also, samples of the antenna **220** were prepared by changing the T shape of the opening **222** of the antenna **220**, and measurements were conducted per sample, to determine the T shape of the opening **222** whereby good communication characteristics could be achieved regardless of the presence or absence of liquid.
(42) As a result of this, good communication characteristics were achieved, regardless of the presence or absence of liquid inside the blood collection tube **10**, when the width W2 of the first opening **223** was 21.60 mm and the depth D2 was 4.00 mm, and the width W3 of the second opening **224** was 4.20 mm and the depth D3 was 6.00 mm. Note that the width D4 of the conductor connecting the IC chip **210** was made 1.00 mm. Also, aluminum (10-μm thick) was used as the conductor to form the antenna **220**.
(43) FIG. **4** is a diagram that illustrates an example of impedance characteristics of the antenna according to one embodiment. FIG. **4**(A) shows a simulation result of impedance characteristics of the antenna **220** when the blood collection tube **10** holds no liquid. Also, FIG. **4**(B) shows a simulation result of impedance characteristics of the antenna **220** when the blood collection tube **10** holds liquid (water).
(44) In FIGS. **4**(A) and (B), the impedance Z of the antenna **220** is “Z=R+jX,” and the vertical axis is the value of the real part R and the value of the imaginary part X of the impedance Z. Also, the horizontal axis is the frequency. Note that, in both graphs, the solid line plots the real part R corresponding to each frequency, and the dashed line plots the imaginary part X corresponding to each frequency.
(45) The example of FIG. **4**(A) shows that, at 920 MHz, which is the center frequency of the frequency band that the RFID tag **100** uses for communication, the value of the real part R in the impedance Z of the antenna **220** is approximately 4.4Ω, and the value of the imaginary part X is approximately 213.0Ω. In the example of FIG. **4**(B), at a frequency of 920 MHz, the value of the real part R in the impedance Z of the antenna **220** is approximately 305.0Ω and the value of the imaginary part X is approximately 236.0Ω. Thus, the impedance Z of the antenna **220** changes depending on whether the blood collection tube **10** holds liquid inside.
(46) Also, from the results of preparing a number of samples of antennas, it was found out that good communication characteristics could be achieved when the value of the imaginary part X at a frequency of 920 MHz was around 213Ω (for example, approximately 190Ω to 240Ω). Based on this, it is likely that, when the value of the imaginary part X of the impedance Z of the antenna is around 213Ω, the internal capacitance of the IC chip **210** and the inductance component of the antenna **220** resonate in the 920 MHz band, and impedance matching is achieved.

(47) That is, FIGS. **4**(A) and (B) show that the value of the imaginary part X of the impedance Z is 190Ω to 240Ω at 920 MHz, so that the antenna **220** achieves good communication characteristics regardless of the presence or absence of liquid in the blood collection tube **10**. Note that the evaluation results of communication characteristics of the RFID tag **100** having the antenna **220** will be described later.

Comparative Example

(48) FIG. **5** is a diagram that illustrates an example of the shape of an antenna according to a comparative example. The antenna **500** shown in FIG. **5** has the same outer shape (W1=35.00 mm, D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **3**, and an opening **502** is formed as a rectangle inside the inner periphery **501** of the antenna **500**. Also, the antenna **500** is formed such that the area of the opening **502** is the same as the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**.

(49) FIG. **6** is a diagram that illustrates an example of impedance characteristics of an antenna according to a comparative example. FIG. **6**(A) shows a simulation result of impedance characteristics of the antenna **500** when the blood collection tube **10** holds no liquid. The example of FIG. **6**(A) shows that, at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **500** is approximately 3.8Ω and the value of the imaginary part X is approximately 204.7Ω. Thus, when the blood collection tube **10** holds no liquid, the impedance Z of the antenna **500** according to the comparative example shows impedance characteristics close to those of the antenna **220** according to one embodiment shown in FIG. **3**.

(50) On the other hand, FIG. **6**(B) shows a simulation result of impedance characteristics of the antenna **500** when the blood collection tube **10** holds liquid (water). The example of FIG. **6**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance Z of the antenna **500** is approximately 164.8Ω, and the value of the imaginary part X is approximately 285.5Ω. Thus, it is clear that the impedance Z of the antenna **500** according to the comparative example changes up to approximately 285Ω when the blood collection tube **10** holds liquid.

(51) Thus, the antenna **220** according to one embodiment shown in FIG. **3** shows a lower impedance Z than the antenna **500** according to a comparative example shown in FIG. **5**, and the value of the imaginary part X, that is, the resonance frequency, changes less, especially when the blood collection tube **10** holds liquid.

Modifications

(52) Next, changes in impedance characteristics when the shape of the T-shaped opening **222** of the antenna **220** is varied will be described by illustrating a number of modifications.

First Modification

(53) FIG. **7** is a diagram that illustrates an example of the shape of an antenna according to a first modification. An antenna **700** according to the first modification shown in FIG. **7** has the same outer diameter (W1=35.00 mm and D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **7**. Also, the antenna **700** is formed such that the area of the opening **702** inside the inner periphery **701** is larger than the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**.

(54) FIG. **8** is a diagram that illustrates an example of impedance characteristics of the antenna according to the first modification. FIG. **8**(A) shows a simulation result of impedance characteristics of the antenna **700** when the blood collection tube **10** holds no liquid. Also, FIG. **8**(B) shows a simulation result of impedance characteristics of the antenna **700** when the blood collection tube **10** holds liquid (water).

(55) The example of FIG. **8**(A) shows that, at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **700** is approximately 10.7Ω, and the value of the imaginary part X is approximately 332.0Ω. The example of FIG. **8**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance

Z of the antenna **700** is approximately 442.0Ω, and the value of the imaginary part X is approximately −139.0Ω.
(56) Also, in FIG. **8**(A), the frequency at which the value of the imaginary part X of the impedance Z of the antenna **700** is around 213Ω changes to a low frequency near 700 MHz.
(57) Thus, by increasing the inner area of the T shape of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**, the resonance frequency of the antenna **220** can be changed (adjusted) to be lower.
Second Modification
(58) FIG. **9** is a diagram that illustrates an example of the shape of an antenna according to a second modification. An antenna **900** according to the second modification shown in FIG. **9** has the same outer shape (W1=35.00 mm and D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **3**. Also, the antenna **900** is formed such that the area of the opening **902** inside the inner periphery **901** is smaller than the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**.
(59) FIG. **10** is a diagram that illustrates an example of impedance characteristics of the antenna according to the second modification. FIG. **10**(A) shows a simulation result of impedance characteristics of the antenna **900** when the blood collection tube **10** holds no liquid. Also, FIG. **10**(B) shows a simulation result of impedance characteristics of the antenna **900** when the blood collection tube **10** holds liquid (water).
(60) In the example of FIG. **10**(A), at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **900** is approximately 2.6Ω, and the value of the imaginary part X is approximately 141.7Ω. The example of FIG. **10**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance Z of the antenna **900** is approximately 171.9Ω, and the value of the imaginary part X is approximately 205.0Ω.
(61) Also, in FIG. **10**(A), the frequency at which the value of the imaginary part X of the impedance Z of the antenna **900** is around 213Ω changes to a high frequency near 1150 MHz.
(62) Thus, by reducing the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**, the resonance frequency of the antenna **220** can be changed (adjusted) to be higher.
Third Modification
(63) FIG. **11** is a diagram that illustrates an example of the shape of an antenna according to a third modification. An antenna **1100** according to the third modification shown in FIG. **11** has the same outer shape (W1=35.00 mm and D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **11**. Also, the antenna **1100** according to the third modification is formed such that the size (area) of the two rectangular parts **1102** and **1103** shown in FIG. **11** is changed so as to make the area of the inner opening **1104** of the inner periphery **1101** larger than the area of the opening **222**. Note that the dashed two-dotted lines in FIG. **11** are imaginary lines for explaining the positions of the rectangular parts **1102** and **1103** and do not really exist.
(64) FIG. **12** is a diagram that illustrates an example of impedance characteristics of an antenna according to the third modification. FIG. **12**(A) shows a simulation result of impedance characteristics of an antenna **1100** when the blood collection tube **10** holds no liquid. Also, FIG. **12**(B) shows a simulation result of impedance characteristics of the antenna **1100** when the blood collection tube **10** holds liquid (water).
(65) The example of FIG. **12**(A) shows that, at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **1100** is approximately 5.54Ω, and the value of the imaginary part X is approximately 236.2Ω. The example of FIG. **12**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance Z of the antenna **1100** is approximately 392.3Ω, and the value of the imaginary part X is approximately 181.4Ω.

(66) Also, FIG. **12**(A) makes it clear that the frequency at which the value of the imaginary part X of the impedance Z of the antenna **1100** is around 213Ω is near 870 MHz.
(67) As described above, when increasing the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**, the size (area) of the rectangular parts **1102** and **1103** shown in FIG. **1**I may be reduced, so that the impedance characteristics of the antenna **220** can be fine-tuned.
Fourth Modification
(68) FIG. **13** is a diagram that illustrates an example of the shape of an antenna according to a fourth modification. An antenna **1300** according to the fourth modification shown in FIG. **13** has the same outer shape (W1=35.00 mm and D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **3**. Also, the antenna **1300** according to the fourth modification is formed such that the size (area) of the two rectangular parts **1102** and **1103** is changed so as to make the area of the opening **1302** inside the inner periphery **1301** smaller than the area of the opening **222** of the antenna **220**.
(69) FIG. **14** is a diagram that illustrates an example of impedance characteristics of the antenna according to the fourth modification. FIG. **14**(A) shows a simulation result of impedance characteristics of the antenna **1300** when the blood collection tube **10** holds no liquid. Also, FIG. **14**(B) shows simulation results of impedance characteristics of the antenna **1300** when the blood collection tube **10** holds liquid.
(70) The example of FIG. **14**(A) shows that, at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **1300** is approximately 3.45Ω, and the value of the imaginary part X is approximately 193.0Ω. The example of FIG. **14**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance Z of the antenna **1300** is approximately 210.6Ω, and the value of the imaginary part X is approximately 254.2Ω.
(71) Also, FIG. **14**(A) makes it clear that the frequency at which the value of the imaginary part X of the impedance Z of the antenna **1300** is around 213 Ω is near 950 MHz.
(72) As described above, when reducing the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3**, the size (area) of the rectangular parts **1102** and **1103** may be increased, so that the impedance characteristics of the antenna **220** can be fine-tuned.
Fifth Modification
(73) FIG. **15** is a diagram that illustrates an example of the shape of an antenna according to a fifth modification. An antenna **1500** according to the fifth modification shown in FIG. **15** has the same outer shape (W1=35.00 mm and D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **3**. Also, with the antenna **1500** according to the fifth modification, when the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3** is increased, the area of the opening **1502** inside the inner periphery **1501** is increased, while maintaining a similar shape (similarity) to the shape of the letter T.
(74) FIG. **16** is a diagram that illustrates an example of impedance characteristics of the antenna according to the fifth modification. FIG. **16**(A) shows a simulation result of impedance characteristics of the antenna **1500** when the blood collection tube **10** holds no liquid. Also, FIG. **16**(B) shows a simulation result of impedance characteristics of the antenna **1500** when the blood collection tube **10** holds liquid (water).
(75) The example of FIG. **16**(A) shows that, at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **1500** is approximately 0.93Ω, and the value of the imaginary part X is approximately 287.0Ω. The example of FIG. **16**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance Z of the antenna **1500** is approximately 492.4Ω, and the value of the imaginary part X is approximately 37.7Ω.
(76) Also, FIG. **16**(A) makes it clear that the frequency at which the value of the imaginary part X

of the impedance Z of the antenna **1500** is around 213Ω is near 750 MHz.
(77) As described above, when the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3** is increased, the size (area) of the opening **222** may be increased, while maintaining a similar shape (similarity) to the shape of the letter T, so that the impedance characteristics of the antenna **220** can be changed more dynamically.
Sixth Modification
(78) FIG. **17** is a diagram that illustrates an example of the shape of an antenna according to a sixth modification. An antenna **1700** according to the sixth modification shown in FIG. **17** has the same outer shape (W1=35.00 mm and D1=16.00 mm) as the antenna **220** according to one embodiment shown in FIG. **3**. Also, with the antenna **1700** according to the sixth modification, when the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3** is reduced, the area of the opening **1702** inside the inner periphery **1701** is reduced while maintaining a similar shape (similarity) to the shape of the letter T.
(79) FIG. **18** is a diagram that illustrates an example of impedance characteristics of the antenna according to the sixth modification. FIG. **18**(A) shows a simulation result of impedance characteristics of the antenna **1700** when the blood collection tube **10** holds no liquid. Also, FIG. **18**(B) shows a simulation result of impedance characteristics of the antenna **1700** when the blood collection tube **10** holds liquid (water).
(80) The example of FIG. **18**(A) shows that, at 920 MHz, which is the frequency the RFID tag **100** uses for communication, the value of the real part R of the impedance Z of the antenna **1700** is approximately 1.79Ω, and the value of the imaginary part X is approximately 137.3Ω. The example of FIG. **18**(B) shows that, at a frequency of 920 MHz, the value of the real part R of the impedance Z of the antenna **1700** is approximately 66.0Ω, and the value of the imaginary part X is approximately 181.7Ω.
(81) Also, in FIG. **18**(A), the frequency at which the value of the imaginary part X of the impedance Z of the antenna **1800** is around 213Ω is estimated to be 1200 MHz or higher.
(82) Thus, the area of the opening **222** of the antenna **220** according to one embodiment shown in FIG. **3** is reduced, and the area of the opening **222** is reduced while maintaining a similar shape (similarity) to the shape of the letter T, so that the impedance characteristics of the antenna **220** can be changed more dynamically.
(83) From the above, it is likely that, with the antenna **220** according to one embodiment shown in FIG. **3**, the resonance frequency of the antenna **220** can be adjusted by varying the inner area of the T shape in the inner periphery **221**.
(84) Also, with the antenna **220**, it is likely that the impedance of the antenna **220**, in particular the impedance when the blood collection tube **10** holds liquid, can be adjusted by varying the T shape of the inner periphery **221**.
(85) <Evaluation Result of RFID Tag>
(86) Next, an evaluation result of the reading characteristics of identification information (tag ID), observed when applying the RFID tag **100** having the antenna **220** according to one embodiment shown in FIG. **3** to the blood collection tube **10**, will be described.
(87) FIG. **19** is a diagram that illustrates an example arrangement pattern of blood collection tubes according to one embodiment. Here, as shown in FIGS. **19**(A) to (D), a total of 100 blood collection tubes **10** were arranged in 10 columns and 10 rows and set in a tag ID reader device, read 100 times, and evaluated by counting the number of times all of the 100 tag IDs were read successfully.
(88) Note that, for the blood collection tubes **10**, blood collection tubes **10** made of plastic (polyethylene terephthalate) were used. Also, the RFID tag **100** was applied to each blood collection tube **10** such that, as shown in FIG. **2**, the plus X direction of the RFID tag **100** faces the cap **11**, and the longitudinal direction of the blood collection tube **10** and the longitudinal direction of the RFID tag **100** were substantially parallel.

(89) Here, shown in FIG. **19**(A), an arrangement pattern, in which four blood collection tubes **10** are grouped as one set, and in which each blood collection tube **10** is arranged such that the RFID tag **100** faces the inner side of the four blood collection tubes **10**, is called “arrangement pattern 1.” Also, shown in FIG. **19**(B), an arrangement pattern, in which the RFID tag **100** of each blood collection tube **10** is arranged to face one direction (for example, forward), is called “arrangement pattern 2.” Furthermore, shown in FIG. **19**(C), an arrangement pattern, in which the RFID tags **100** of the blood collection tubes **10** of the outermost row are arranged to face outward, and in which the other blood collection tubes **10** are arranged as shown in FIG. **19**(C), is called “arrangement pattern 3.” Also, shown in FIG. **19**(D), an arrangement pattern, in which the RFID tags **100** of the blood collection tubes **10** of the outermost row are arranged to face inward, and in which the other blood collection tubes **10** are arranged as shown in FIG. **19**(D), is called “arrangement pattern 4.”
(90) FIG. **20** is a diagram that illustrates an evaluation result of RFID tags according to one embodiment. As for the detail of evaluation, RFID tags **100** with the antenna **220** according to one embodiment shown in FIG. **3** were applied to blood collection tubes **10**, and, the tag IDs were read by a reader device, 100 times, in each arrangement pattern of arrangement patterns 1 to 4. Note that the transmit/output power level of the reader device was 19 dBm.
(91) Also, as a reference, the same reading was performed with blood collection tubes to which an existing RFID tag was applied.
(92) FIG. **20**(A) shows an evaluation result, in which the number of times all the tag IDs (100 tag IDs) were read successfully when the blood collection tubes **10** were empty (holding no liquid) and the tag IDs were read 100 times in each of arrangement patterns 1 to 4 was recorded.
(93) As shown in FIG. **20**(A), with the blood collection tubes **10**, to which the RFID tag **100** according to the present embodiment was applied, all the tag IDs were read successfully, 100 times, in all of arrangement patterns 1 to 4. On the other hand, with the blood collection tubes to which an existing RFID tag was applied, reading failures occurred in arrangement patterns 1, 3, and 4, and, especially in arrangement pattern 1, the number of times all the tag IDs were read successfully was 0.
(94) FIG. **20**(B) shows an evaluation result, in which the number of times all the tag IDs (100 tag IDs) were read successfully when the blood collection tube **10** contained water (an example of liquid) and the tag IDs were read 100 times in each of arrangement patterns 1 to 4 was recorded.
(95) As shown in FIG. **20**(B), with the blood collection tubes **10**, to which the RFID tag **100** according to the present embodiment was applied, all the tag IDs were read successfully, 100 times, in all of arrangement patterns 1 to 4. On the other hand, with the blood collection tubes to which existing RFID tags were applied, all the tag IDs were read successfully, 100 times, in arrangement patterns 2 to 4, but in arrangement pattern 1, all the tag IDs (100 tag IDs) were read successfully only four times.
(96) As described above, according to the present embodiment, the RFID **100** tag applied to a container holding liquid such as the blood collection tube **10** can achieve a good communication distance regardless of the presence or absence of liquid.
(97) Also, the antenna **220** included in the RFID tag **100** according to the present embodiment can change the resonance frequency of the antenna **220** by changing the area of the opening, without changing the outer shape, and therefore is suitable for small containers with limitations on their outer shape.
(98) Furthermore, the antenna **220** according to the present embodiment can, for example, change the shape of the opening (to a dissimilar shape) while maintaining the area of the opening **222** where no conductor is formed, thereby making it easy to adjust the impedance characteristics of the antenna **220** when liquid is held.
(99) As described above, according to the embodiment of the present invention, it is possible to provide an RFID tag **100** that is applied to a container for holding liquid and that can achieve a good communication distance regardless of the presence or absence of liquid.

(100) The structure shown in the above embodiment only illustrates an example of the present invention, and it is possible to combine the present invention with other known technologies, and furthermore, part of the structure can be omitted or changed without departing from the scope of the present invention.
(101) This application is based on and claims priority to Japanese Patent Application No. 2020-197057, filed with Japan Patent Office on Nov. 27, 2020, the entire contents of which are incorporated herein by reference.
DESCRIPTION OF REFERENCE NUMERALS
(102) **10** blood collection tube (an example of a container) **100** RFID tag **210** IC chip **220** antenna **221** inner periphery of antenna **222** opening **223** first opening **224** second opening

Claims

1. An RFID (radio frequency identification) tag adapted to be applied to a container for holding liquid, the RFID tag comprising: an IC (integrated circuit) chip on which identification information is recorded; and an antenna made of a loop-shaped conductor connected to the IC chip, wherein the antenna has a T-shaped opening in which the conductor is not formed, wherein an outer shape of the antenna forms a rectangle, wherein the IC chip is electrically connected to a central portion of one side of the rectangle, and wherein the T-shaped opening includes: a first opening extending in a direction parallel to the one side of the rectangle, and a second opening connected to a central part of the first opening, extending in a direction orthogonal to the one side of the rectangle, and extending in a direction opposite to the one side.
2. The RFID tag according to claim 1, wherein a resonance frequency of the antenna is adjusted by varying an area of the T-shaped opening.
3. The RFID tag according to claim 1, wherein an impedance of the antenna is adjusted by varying a shape of the T-shaped opening.
4. The RFID tag according to claim 3, wherein an impedance of the antenna observed when the container holds liquid therein is adjusted by varying a shape of the T-shaped opening.
5. The RFID tag according to claim 1, wherein the RFID tag uses frequencies including a 920 MHz band, between 860 MHz and 960 MHz, for communication.
6. The RFID tag according to claim 1, wherein the container is a blood collection tube configured to hold blood.
7. A blood collection tube with the RFID tag according to claim 1 applied thereto.
8. An antenna for use for an RFID (radio frequency identification) tag adapted to be applied to a container for holding liquid, wherein the antenna is adapted to be connected to an IC (integrated circuit) chip; wherein the antenna is made of a loop-shaped conductor, wherein the loop-shaped conductor has a T-shaped opening in which the conductor is not formed, wherein an outer shape of the antenna forms a rectangle, wherein a central portion of one side of the rectangle is adapted to be electrically connected to the IC chip, and wherein the T-shaped opening includes: a first opening extending in a direction parallel to the one side of the rectangle, and a second opening connected to a central part of the first opening, extending in a direction orthogonal to the one side of the rectangle, and extending in a direction opposite to the one side.